Laszlo Edgar Dobra

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EDUCATION

University of Illinois Urbana-Champaign Expected December 2026

Bachelor of Science in Chemical Engineering, Minor in Semiconductor Engineering **GPA: 3.97**

- **Honors:** Dean's List, IL Semiconductor Workforce Network Fellow, IL Engineering Achievement & Outstanding Scholar
- **Coursework:** Reaction Engineering, Condensed Matter Physics, Electronic Properties of Materials, Thermodynamics, Process Control

EXPERIENCE

Rust-Oleum June 2025 – August 2025

R&D Intern

- Spearheaded formulation of silicate-based coatings, establishing baseline fluid processes for industrial scale-up to penetrate a **$12B+ global market**.
- Engineered a **90% reduction in fluid viscosity** via advanced rotational rheometry, optimizing flow dynamics for large-scale material dispensing.
- Modeled ionic interactions to resolve thermal curing anomalies, improving process yield and long-term formulation stability.
- Validated long-term material reliability by designing rigorous accelerated weathering (QUV) and ASTM testing protocols.
- Presented R&D process breakthroughs and scale-up strategies to **40+ senior executives and lead scientists**.

Guironnet Research Group October 2023 – May 2026

Research Assistant

- Designed liquid handler robotics for automated high-throughput synthesis, establishing robust process control for the precise production of **100+ polymers**.
- Engineered and operated specialized air- and moisture-sensitive environments utilizing gloveboxes, pressurized gas, and **high-vacuum systems**.
- Developed a GPC metrology deconvolution algorithm (Python, NumPy) deployed as a Streamlit web app, **accelerating data analysis by 50x**.
- Showcased automated synthesis and process breakthroughs at 5 major conferences, securing a highly competitive award against **30+ top researchers**.

PROJECTS

MEMS and NEMS Cleanroom Fabrication January 2025 – May 2025

Course-Based Lab Project

- Engineered pressure sensors on silicon nitride coated wafers via end-to-end cleanroom processing (**photolithography, PVD, wet/dry etching**).
- Validated nanoscale tolerances and critical dimensions utilizing profilometry and direct electrical probing.
- Architected PDMS microfluidic devices using Fusion 360 and K-Layout, demonstrating optimized microscale fluid dynamics.

Chemical Reaction Engineering Design April 2026

Reactor Modeling Project

- Modeled a methane-to-ethane packed bed reactor with a recycle loop, evaluating process dynamics, chemical kinetics, and mass transfer.
- Developed Python simulation scripts (SciPy, Matplotlib) to optimize thermal management parameters for high-yield conversion.

SKILLS

- **Semiconductor & Cleanroom:** Photolithography (UV), CVD, PVD, Wet/Dry Etching, Profilometry, High-Vacuum Systems, K-Layout
- **Metrology & Analysis:** GPC, HPLC, GC, NMR, Rheology, Nanoscale Thermometry, UV-Vis & IR Spectroscopy
- **Programming:** Python (NumPy, SciPy, Matplotlib, Streamlit), MATLAB, C, JavaScript, HTML
- **Software:** SAP, Fusion 360, Siemens NX, ChemDraw, Microsoft Office, Adobe Creative Cloud
- **Languages:** English (Fluent), Hungarian (Fluent), Romanian (Fluent), German (Intermediate)